

HIMANSHU NAKRANI

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PROFESSIONAL SUMMARY

AI Engineer with over 2 years of experience architecting and deploying production-grade Generative AI and NLP applications, agentic AI systems, and RAG pipelines. Proven expertise in fine-tuning large language models, developing sophisticated Text-to-SQL systems, and engineering scalable backend infrastructures that drive significant performance and efficiency improvements for enterprise data solutions.

TECHNICAL SKILLS

Generative AI & LLMs: Large Language Models, RAG (Retrieval-Augmented Generation), Agentic AI, Multi-agent Systems, AI Guardrails, Text-to-SQL, Fine-tuning (LoRA/PEFT), Prompt Engineering, Embeddings, Model Optimization, LangChain, Hugging Face, OpenAI API, Azure OpenAI, Google ADK

Backend Development: FastAPI, ASP.NET Core, REST APIs, SQLAlchemy, Microservices, API Design, OOP

Databases: SQL, Vector Databases (pgvector), Semantic Search, Query Optimization

Machine Learning: Deep Learning, NLP, Transformers, PyTorch, Model Deployment, Data Augmentation

Cloud Platforms: Azure, AWS, Oracle Cloud Infrastructure (OCI), Databricks, Azure OpenAI Studio

Programming Languages: Python, Java, SQL, C/C++, C#, Shell/Perl Scripting

Developer Tools: Git, Docker, Harness, CI/CD, MLOps, LLMops, Model Observability, Testing, Evaluation Frameworks

PROFESSIONAL EXPERIENCE

State Street Corporation

Hyderabad, India

Senior Associate - AI Software Developer

July 2023 – Present

- Architected and deployed "**Alpha Copilot**", an enterprise LLM-powered Text-to-SQL chatbot enabling natural language queries over structured financial data, serving 100+ internal users
- Engineered scalable AI inference infrastructure using **FastAPI** and **SQLAlchemy**, implementing advanced caching and **latency optimization** techniques that reduced average response time by **75% (from 25-30s to 6-8s)**
- Developed production-ready **RAG system** for extracting actionable insights from fund prospectuses, engineering an optimized document pipeline with intelligent chunking, semantic embeddings, and vector database integration achieving sub-second retrieval times
- Integrated LLM-based data visualization capabilities, automatically generating interactive charts and summaries from query results, improving data accessibility and decision-making speed
- Led backend development for **WealthAI application**, building real-time portfolio narrative generation system that processes optimizer outputs and delivers personalized investment insights at scale
- Fine-tuned **GPT-4o mini**, **GPT-4.1 mini** (Azure OpenAI) and **Llama 3.1-8B** (Databricks) using supervised fine-tuning and PEFT/LoRA, managing end-to-end LLMops pipeline including data augmentation, training, and inference optimization
- Conducted research on advanced Text-to-SQL generation techniques, exploring schema-linking methods and few-shot prompting strategies to improve SQL accuracy by 25%
- Spearheaded development of "**Agent Forge**", an enterprise **agentic AI** platform centralizing agent registration, discovery, and governance — providing a unified inventory of metadata, capabilities, ownership, and performance indicators across **multi-agent systems** in production
- Designed and implemented OTEL-compliant **model observability** and automated evaluation pipelines for real-time agent behavior analysis and **AI guardrails**; currently building a no-code agentic workflow builder for visually designing and deploying custom AI agents
- Achieved **95%+ unit test coverage** and streamlined deployment workflows by architecting automated **CI/CD pipelines** with **Harness** and **Docker**, significantly improving system reliability and reducing production incidents
- Collaborated with cross-functional teams in an Agile environment, partnering with business analysts and compliance officers to align deliverables with regulatory requirements and business objectives

State Street Corporation

Hyderabad, India

Software Development Intern

January 2023 – June 2023

- Led migration of legacy ideation platform from .NET Core to modern **FastAPI + ReactJS** stack, improving application performance by 40% and developer productivity
- Designed and implemented RESTful APIs with SQLite backend, ensuring seamless frontend integration and comprehensive test coverage

Technocolabs Software Inc.

Remote

Machine Learning Developer Intern

June 2022 – July 2022

- Developed and deployed CNN-based **Spotify Skip Action Predictor**, achieving 87% accuracy in real-time song skip prediction
- Built end-to-end ML pipeline for audio classification tasks including data preprocessing, feature engineering, model training, and API deployment

PUBLICATIONS & RESEARCH

GoT4SQL-DA: A Graph-of-Thoughts Framework for Scalable Text-to-SQL Data Augmentation	FLLM 2025
GRAFT: Graph-of-Thoughts based Reasoning and Augmentation for Finetuning Text-to-SQL	CSCI 2025

EDUCATION

Nirma University	Ahmedabad, India
Bachelor of Technology in Electronics and Communication Engineering	June 2019 – July 2023

KEY PROJECTS

CogniDoc: RAG Document Q&A Platform <i>FastAPI, Supabase pgvector, OpenAI, Gemini, Vercel</i> Live
• Developed a production-ready Retrieval-Augmented Generation (RAG) pipeline encompassing document ingestion, intelligent chunking, and context-aware semantic retrieval utilizing Supabase pgvector
• Engineered a scalable FastAPI backend with multi-model LLM routing (OpenAI, Gemini) to optimize response latency and relevance, empowering real-time conversational memory in a Vercel -deployed application
Intrusion Detection System <i>Python, Machine Learning, Cybersecurity</i> Kaggle
• Developed ML-based network intrusion detection classifier using KDD Cup 1999 dataset with 4.9M connection records, achieving 94% accuracy in identifying malicious network traffic patterns across multiple attack categories
• Implemented comprehensive feature engineering pipeline and trained ensemble models (Random Forest, XGBoost) for real-time anomaly detection, successfully classifying DoS, Probe, R2L, and U2R attacks
Bitcoin Price Prediction (Daily Updates) <i>Python, Deep Learning, Time Series</i> Kaggle
• Built automated daily-updated cryptocurrency price forecasting system using LSTM networks on historical Bitcoin data from Cryptocurrency Prices Dataset, incorporating technical indicators and market sentiment features
• Engineered sequential model architecture with dropout regularization and achieved 18% RMSE reduction compared to baseline models through hyperparameter tuning and walk-forward validation

CERTIFICATIONS

Oracle Cloud Infrastructure Certified Generative AI Professional View Certificate
Generative AI Fundamentals (Databricks) View Certificate
Amazon Machine Learning Summer School 2022 View Certificate
AWS Machine Learning Foundations (Udacity) View Certificate
Goldman Sachs Engineering Virtual Program Participant View Certificate